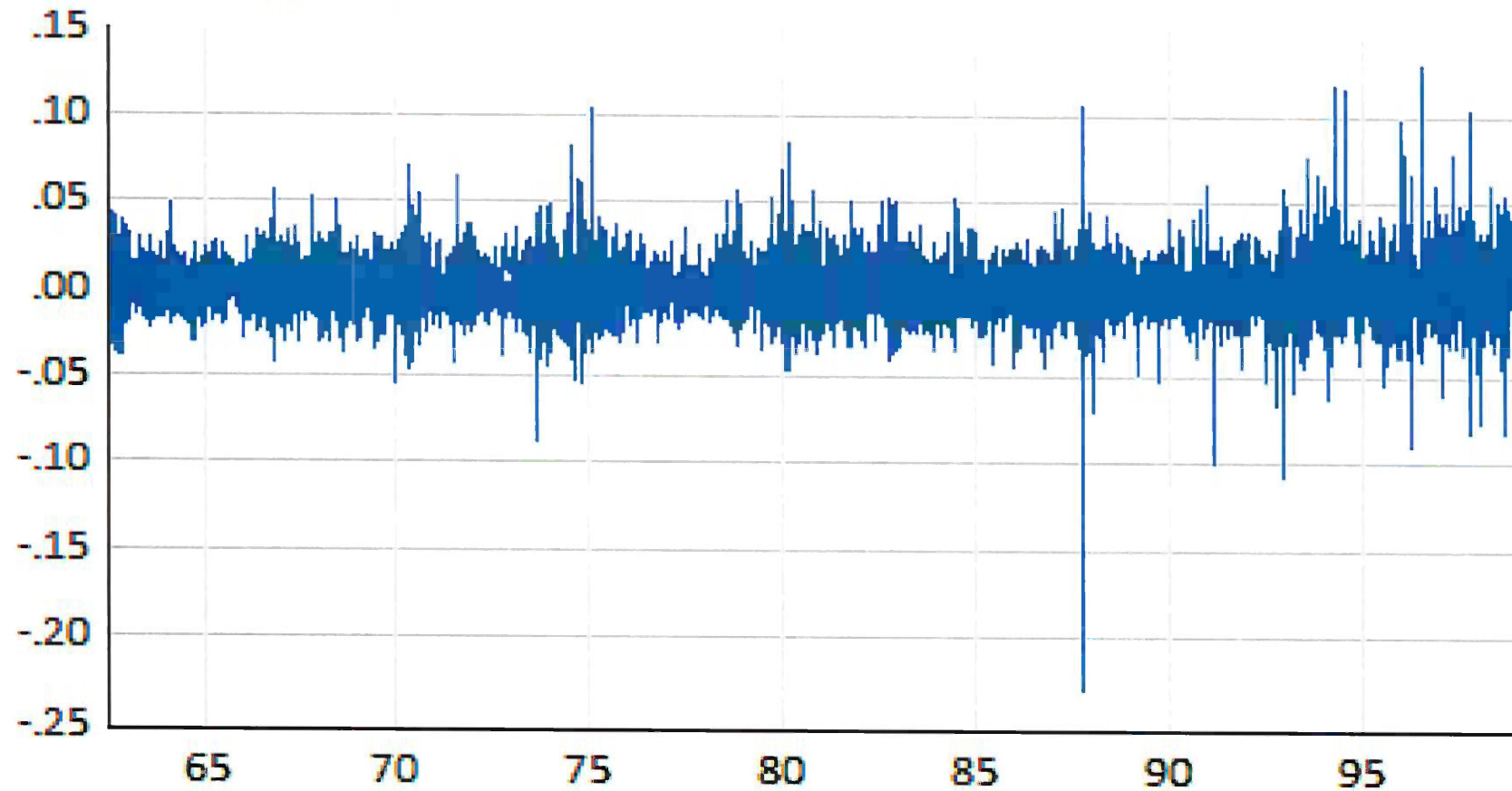


Week 11: IGARCH and ARMA models

Consider the daily log returns of IBM (Y_t) stock from 3 July 1962 to 31 December 1998.

not %



GARCH

Consider the following model:

$$Y_t = c + \varepsilon_t$$

no AR(1) term.

think "c" about zero

and

$$\varepsilon_t = \sigma_t u_t$$

with a GARCH(1,1) model

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

```
.  
> df <- readxl::read_excel("week 11.xlsx")  
>  
> library(xts)  
> rtn.xts = as.xts(df) # create xts object  
> mean(rtn.xts)  
[1] 0.000556556  $\Rightarrow$  sample mean ( $\bar{r}_t$ )
```

```
spec_garch11 <- ugarchspec(  
  variance.model = list(  
    model = "sGARCH",  
    garchOrder = c(1, 1)  
  ),  
  mean.model = list(  
    armaOrder = c(0, 0),  
    include.mean = TRUE  
  ),  
  distribution.model = "norm"  
)  
fit_garch11 <- ugarchfit(  
  spec = spec_garch11,  
  data = rtn.xts  
)  
fit_garch11@fit$matcoef
```

```
> fit_garch11@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
$\hat{C} = \mu$	6.942479e-04	1.326736e-04	5.232750	1.670066e-07
$\hat{\alpha}_0 = \omega$	2.642554e-06	6.413163e-07	4.120515	3.780265e-05
alpha1	6.180209e-02	5.754673e-03	10.739463	0.000000e+00
beta1	9.288583e-01	6.893984e-03	134.734618	0.000000e+00

Wow!

IGARCH Models

The estimated GARCH(1,1) model

$$Y_t = \overset{\text{mu}}{0.0007} + \hat{\varepsilon}_t$$

$$\begin{aligned}\hat{\sigma}_t^2 &= \hat{\alpha}_0 + \hat{\alpha}_1 \hat{\varepsilon}_{t-1}^2 + \hat{\beta}_1 \hat{\sigma}_{t-1}^2 \\ &= \underset{\text{omega}}{0.0000027} + \underset{\text{alpha 1}}{0.062} \hat{\varepsilon}_{t-1}^2 + \underset{\text{beta 1}}{0.929} \hat{\sigma}_{t-1}^2\end{aligned}$$

Note that the fitted model shows $\hat{\alpha}_1 + \hat{\beta}_1 = 0.062 + 0.929 = 0.991$, < 1 (2) which is close to 1.

This phenomenon is commonly observed in practice and it leads to imposing the constraint $\alpha_1 + \beta_1 = 1$ in a GARCH(1,1) model.

① Also notice that $\hat{\alpha}_0$ is close to 0.

IGARCH Models

→ integrated.

If $\alpha_1 + \beta_1 = 1$ then $\alpha_1 = 1 - \beta_1$.

With $\alpha_1 = 1 - \beta_1$ the GARCH(1,1) model is called the **integrated-GARCH** (or **IGARCH**) model:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

$$= \alpha_0 + (1 - \beta_1) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

α_0 and β_1

IGARCH(1,1)
model.

Observe that $0 < \beta_1 < 1$.

IGARCH Models

Recall that the unconditional variance of ε_t for an GARCH(1,1) model is

$$\text{var}(\varepsilon_t) = \frac{\alpha_0}{1 - (\alpha_1 + \beta_1)} = \frac{\alpha_0}{1 - 1} = \frac{\alpha_0}{0} = \text{really?}$$

For an IGARCH(1,1) model, $\alpha_1 + \beta_1 = 1$, and so $\text{var}(\varepsilon_t)$ does not exist.

This seems hard to justify since it implies that the series is not stationary.

$$\alpha_0 = 0 \quad \text{var}(\varepsilon_t) = 0$$

‘Special’ IGARCH Models

(1)

It is of particular interest to consider $\alpha_0 = 0$. This is known as an IGARCH(1,1) model.

$$\sigma_t^2 = (1 - \beta_1) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

only need β_1

R will estimate this IGARCH(1,1) model with $\alpha_0 = 0$.

For illustration, we estimate this special IGARCH(1,1) model for the daily log returns of IBM stock

$$Y_t = c + \varepsilon_t$$

and

$$\sigma_t^2 = (1 - \beta_1) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

```
spec_special_igarch <- ugarchspec(
  variance.model = list(
    model = "iGARCH",
    garchOrder = c(1, 1)
  ),
  mean.model      = list(
    armaOrder = c(0, 0),
    include.mean = TRUE
  ),
  distribution.model = "norm",

  fixed.pars = list(omega = 0)  ⇒  $\alpha_0 = 0$ 
)
```

```
fit_special_igarch <- ugarchfit(
  spec = spec_special_igarch,
  data = rtn.xts
)
```

```
fit_special_igarch@fit$matcoef
```

```
> fit_special_igarch@fit$matcoef
```

		Estimate	Std. Error	t value	Pr(> t)
$\hat{c}$ $\Leftarrow$ mu		0.0007008544	0.0001343757	5.215635	1.831884e-07
	omega	0.0000000000	NA	NA	NA
	alpha1	0.0392944376	0.0029877176	13.151992	0.000000e+00
$\hat{\beta}_1$ $\Leftarrow$ beta1		0.9607055624	NA	NA	NA

$\Downarrow$ FYI $\Rightarrow \alpha_1 + \beta_1 = 1$

IGARCH

The estimated IGARCH(1,1) model is

$$Y_t = \overset{\text{mu}}{0.000701} + \hat{\varepsilon}_t$$

and

$$\hat{\sigma}_t^2 = (1 - \overset{\text{beta1}}{0.960706}) \hat{\varepsilon}_{t-1}^2 + \overset{\text{beta1}}{0.960706} \hat{\sigma}_{t-1}^2$$

J. P. Morgan RiskMetrics

J. P. Morgan developed the RiskMetrics method to VaR calculation.

RiskMetrics assumes that the daily log return (r_t) follows a conditional normal distribution; that is

$$r_t|F_{t-1} \sim N(\mu_t, \sigma_t^2)$$

↳ conditional $\Rightarrow$ having observed

F_{t-1} is the information set available at time $t - 1$

$\mu_t = E(r_t|F_{t-1})$ is the conditional mean of r_t

σ_t^2 is the conditional variance of r_t

VaR

If we assume that

not %
↑

$$r_t | F_{t-1} \sim N(\mu_t, \sigma_t^2)$$

the VaR_α can be calculated as follows:

Step 1:

Calculate q_α as

$\mu + \sigma \Rightarrow \text{week 1}$

$$q_\alpha = \mu_{T+1|T} + \sigma_{T+1} z_\alpha$$

where $\mu_{T+1|T}$ is the 1-step ahead forecast of the conditional mean

$$\mu_{T+1|T} = E(r_{T+1} | F_T)$$

and σ_{T+1}^2 is the 1-step ahead forecast of the conditional variance.

Week 1: $q_\alpha = \mu + \sigma z_\alpha$

VaR

Step 2:

The $\alpha \times 100\%$ VaR is defined as

$$\text{VaR}_\alpha = |W_0 \times q_\alpha|$$

RiskMetrics

J. P. Morgan RiskMetrics assumes that the following two quantities evolve over time according to the simple model:

$$\mu_t = 0$$

Assume ① $\mu_t = E(r_t | F_{t-1})$

and

$$\sigma_t^2 = \beta_1 \sigma_{t-1}^2 + (1 - \beta_1) r_{t-1}^2$$

with $0 < \beta_1 < 1$.

Thus RiskMetrics assumes that the conditional mean is zero and σ_t^2 follows an IGARCH(1,1) process.

$\equiv 0$
 $\downarrow$
why?
 coz
 sample
 mean about
 zero.

only need β_1 estimate.

$\downarrow$
wow!

RiskMetrics

Specifically RiskMetrics assumes that r_t evolves according to the following model

$$r_t = \varepsilon_t$$

$$r_t = \begin{matrix} c \\ 1 \\ 0 \end{matrix} + \varepsilon_t$$

and

$$\varepsilon_t = \sigma_t u_t, \quad u_t \sim i.i.d.N(0, 1)$$

$$\sigma_t^2 = \overset{\text{wow}}{\beta_1 \sigma_{t-1}^2} + (1 - \beta_1) \underbrace{r_{t-1}^2}_{\varepsilon_{t-1}^2}$$

$$\alpha_i + \beta_i = 1$$

observed slide 2

When $t = T + 1$, we have

$$\sigma_{T+1}^2 = \beta_1 \sigma_T^2 + (1 - \beta_1) r_T^2.$$

↓
conditional
variance.

↓
today return - squared.

J. P. Morgan RiskMetrics

We use daily log returns of IBM stock (denoted as r_t) to illustrate the calculation of VaR.

Question: What is the 5% VaR over the next trading day on the \$10,000,000 investment in IBM stock using the J. P. Morgan RiskMetrics method?

```

spec_JP_risk <- ugarchspec(
  variance.model = list(
    model = "iGARCH",
    garchorder = c(1, 1)
  ),
  mean.model = list(
    armaorder = c(0, 0),  $\Rightarrow$  no AR
    include.mean = FALSE  $\Rightarrow$   $C = 0$  before "true"
  ),
  distribution.model = "norm",

  fixed.pars = list(omega = 0)  $\alpha_0 = 0$ 
)

fit_JP_risk <- ugarchfit(
  spec = spec_JP_risk,
  data = rtn.xts
)

fit_JP_risk@fit$matcoef

```

```
> fit_JP_risk@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
omega	0.00000000	NA	NA	NA
alpha1	0.03816874	0.002920518	13.06917	0
beta1	0.96183126	NA	NA	NA

$\hat{\beta}_1 \Rightarrow$ ~~that~~ that's all

$\rightarrow$ FYI $\Rightarrow \hat{\alpha}_1 + \hat{\beta}_1 = 1$

$$\hat{\alpha}_1 = 1 - \hat{\beta}_1$$

RiskMetrics

The fitted IGARCH model is

$$\begin{aligned}\hat{\sigma}_t^2 &= (1 - \hat{\beta}_1) r_{t-1}^2 + \hat{\beta}_1 \hat{\sigma}_{t-1}^2 \\ &= 0.038169 r_{t-1}^2 + 0.961831 \hat{\sigma}_{t-1}^2.\end{aligned}$$

↓
beta1 slide 21

Conditional standard deviation (σ_t) → slide 20
`sigmahat.JP_risk.xts <- sigma(fit_JP_risk)`
 # Conditional variance (σ_t^2)
`sigmahat2.JP_risk.xts <- sigmahat.JP_risk.xts^2`

square

`> tail(sigmahat2.JP_risk.xts)`

[,1]

1998-12-23 0.0004162978

1998-12-24 0.0004090995

1998-12-28 0.0004031099

1998-12-29 0.0003895833

1998-12-30 0.0003795270

1998-12-31 0.0003651936 ⇒ $\hat{\sigma}_T^2$

r_t ⇒ `> tail(rtn.xts)` → slide 3

Ret

1998-12-23 0.01509

1998-12-24 0.01588

1998-12-28 0.00698

1998-12-29 -0.01123

1998-12-30 -0.00200

1998-12-31 -0.01272 ⇒ r_T

→ r_{T-1} → slide 33

1-step

The sample 1-step ahead forecast is

$$\hat{\sigma}_{T+1}^2 = (1 - \hat{\beta}_1) r_T^2 + \hat{\beta}_1 \hat{\sigma}_T^2$$

slide 23

slide 23

At $T = 31/12/1998$, $r_T = -0.01272$ and $\hat{\sigma}_T^2 = 0.000365$. Thus

$$\begin{aligned}\hat{\sigma}_{T+1}^2 &= 0.038169 r_T^2 + 0.961831 \hat{\sigma}_T^2 \\ &= (0.038169) (-0.01272)^2 + 0.961831 (0.000365) \\ &= 0.000357\end{aligned}$$

VaR

What is the 5% VaR over the **next** trading day on the \$10,000,000 investment based on the J. P. Morgan method?

We calculate $q_{0.05}$ as

$$\begin{aligned} q_{0.05} &= \hat{\mu}_{T+1|T} + \hat{\sigma}_{T+1} z_{\alpha} \\ &= 0 + \sqrt{0.000357} (-1.645) \\ &= -0.031 \end{aligned}$$

$$E(r_t | F_{t-1}) = 0$$

Hence the 1-day horizon 5% VaR is

$$\begin{aligned} VaR_{0.05} &= |10,000,000 (-0.031)| \\ &= \$310,000. \end{aligned}$$

AR-GARCH models

The AR(p) models can be used to model the mean equation, and the conditional heteroscedastic models are used to handle the volatility. Suppose r_t follows an AR(p)-GARCH(1,1) model; that is

$$r_t = \mu_t + \varepsilon_t$$

$$\mu_t = c + \phi_1 r_{t-1} + \cdots + \phi_p r_{t-p} = E(r_t | F_{t-1})$$

AR(p)

with

$$\varepsilon_t = \sigma_t u_t \quad u_t \sim i.i.d.N(0, 1)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Example: IBM stock

We now consider the econometric approach to VaR calculation.

Question: What is the 5% VaR over the next trading day on the \$10,000,000 investment in IBM stock using the AR(p)-GARCH(1,1) model?

Example: IBM stock

Consider a **restricted** AR(2)-GARCH(1,1) model

$$r_t = \phi_0 + \phi_2 r_{t-2} + \varepsilon_t$$

and

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

0

"

$\phi_1 \Gamma_{t-1} \Rightarrow$ Assume.

```

spec_ar2garch11 <- ugarchspec(
  variance.model = list(
    model = "sGARCH",
    garchorder = c(1, 1)
  ),
  mean.model = list(
    armaorder = c(2, 0),
    include.mean = TRUE
  ),
  distribution.model = "norm",
  fixed.pars = list(ar1 = 0) ⇒ impose  $\phi_1 = 0$ 
)

fit_ar2garch11 <- ugarchfit(
  spec = spec_ar2garch11,
  data = rtn.xts
)

fit_ar2garch11@fit$matcoef

```

$\mu \Rightarrow \hat{c}$ is statistically significant.

```
> fit_ar2garch11@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	7.000166e-04	1.296130e-04	5.400820	6.633706e-08 hehe
ar1	0.000000e+00	NA	NA	NA
ar2	-2.467268e-02	1.107454e-02	-2.227875	2.588886e-02 hehe
omega	2.698798e-06	6.844514e-07	3.943008	8.046585e-05
alpha1	6.242375e-02	6.011161e-03	10.384641	0.000000e+00
beta1	9.279848e-01	7.243114e-03	128.119588	0.000000e+00

$\hat{\phi}_2$ is
significant.

$$\hat{\alpha}_1 + \hat{\beta}_1 < 1$$

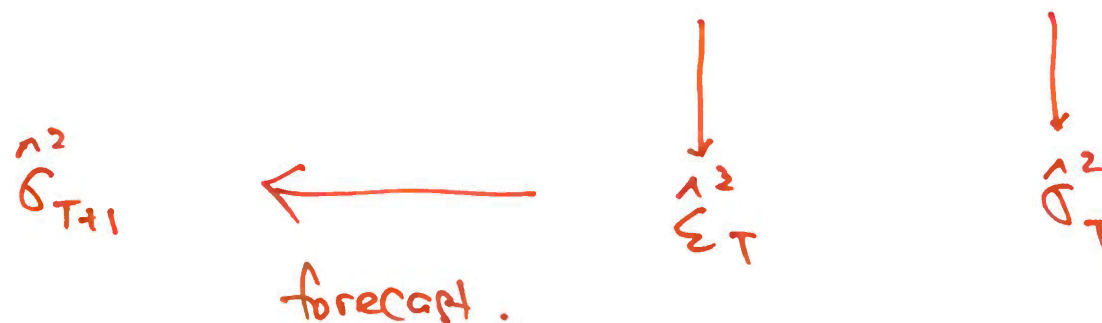
Example: IBM stock

The fitted AR(2)-GARCH(1,1) model is

$$r_t = \overset{\text{mu}}{0.0007} - \overset{\text{ar2}}{0.025}r_{t-2} + \hat{\varepsilon}_t \quad \text{in } \mathbb{R}$$

and

$$\hat{\sigma}_t^2 = \overset{\text{omega}}{0.0000027} + 0.0624\hat{\varepsilon}_{t-1}^2 + 0.9280\hat{\sigma}_{t-1}^2$$



```
> res.ar2garch11.xts <- residuals(fit_ar2garch11)
```

```
> tail(res.ar2garch11.xts)
```

```
          [,1]
```

```
1998-12-23  0.015064781
```

```
1998-12-24  0.015984559
```

```
1998-12-28  0.006635023
```

```
1998-12-29 -0.011555486
```

```
1998-12-30 -0.002545073
```

```
1998-12-31 -0.013714362  $\hat{\varepsilon}_T$  not square
```

```
> # Conditional variance ( $\sigma_t^2$ )
```

```
> sigma2hat.ar2garch11.xts <- sigma(fit_ar2garch11)^2
```

```
> tail(sigma2hat.ar2garch11.xts)
```

```
          [,1]
```

```
1998-12-23 0.0004252821
```

```
1998-12-24 0.0004115211
```

```
1998-12-28 0.0004005338
```

```
1998-12-29 0.0003771362
```

```
1998-12-30 0.0003610108
```

```
1998-12-31 0.0003381157  $\Rightarrow \hat{\sigma}_T^2$ 
```

1-step

The 1-step ahead forecast for the conditional mean $\hat{\mu}_{T+1|T}$ is

μ

$$\begin{aligned}\hat{\mu}_{T+1|T} &= \overset{\text{mu}}{0.0007} - \overset{\text{ar2}}{0.02467}r_{T-1} && \Rightarrow \text{replace "t" with "T+1" in slide 31} \\ &= 0.00071 - 0.02467(-0.002) \\ &= 0.00075 && \downarrow \text{slide 32}\end{aligned}$$

The 1-step ahead forecast for the conditional variance $\hat{\sigma}_{T+1}^2$ is

σ

$$\begin{aligned}\hat{\sigma}_{T+1}^2 &= 0.0000027 + 0.0624\hat{\varepsilon}_T^2 + 0.9280\hat{\sigma}_T^2 \\ &= 0.0000027 + 0.0624(-0.0137)^2 + 0.9280(0.000338) \\ &= 0.00033\end{aligned}$$

$\downarrow \quad \downarrow \quad \rightarrow \text{slide 32}$

VaR

What is the 5% VaR over the next day on the \$10,000,000 investment based on the AR(2)-GARCH(1,1) model?

We calculate $q_{0.05}$ as

$$\begin{aligned} q_{\alpha} &= \overset{\mu}{\hat{\mu}_{T+1|T}} + \overset{\sigma}{\hat{\sigma}_{T+1|T}} z_{\alpha} \\ &= 0.00075 + \sqrt{0.00033} (-1.645) \\ &= -0.0291 \end{aligned}$$

Hence the 1-day-ahead 5% VaR is

$$\begin{aligned} VaR_{0.05} &= |10,000,000 (-0.0291)| \\ &= \$291,000 \end{aligned}$$

VaR

We have applied two methods of VaR calculation to the daily log returns of IBM stock on the \$10 million investment.

The results for the 5% VaR over the next trading day are:

1. \$310,000 for the RiskMetrics
2. \$291,000 for a restricted AR(2)-GARCH(1,1) model.

$\approx \$20,000$

12:54
1:00 back.

ARMA models

Moving Average Model: MA Models

An alternative way to introduce dynamics into a time series model is by including lags of the disturbance term u_t . A first order moving average model is specified as

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

where u_t is a disturbance term distributed as

$$u_t \sim i.i.d.(0, \sigma_u^2)$$

This model is referred to as MA(1).

MA Models

An MA(2) model is specified as

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2}$$

An MA(q) model is specified as

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2} + \cdots + \theta_q u_{t-q}$$

MA

Consider the MA(1) model

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

Unconditional Mean

The unconditional mean of y_t is found by taking (unconditional) expectations of both sides

$$E(y_t) = \phi_0 + E(u_t) + \theta_1 E(u_{t-1})$$

This expression simplifies by using the property of the disturbance term $E(u_t) = E(u_{t-1}) = 0$, in which case the unconditional mean is

$$\mu = E(y_t) = \phi_0$$

Unconditional Variance

Consider the MA(1) model

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

and its “mean” deviation form

$$\mu = \phi_0$$

Subtracting the second equation from the first enables the model to be rewritten in terms of deviations from the mean as

de-mean

$$y_t - \mu = u_t + \theta_1 u_{t-1}$$

so the model is reexpressed in deviations from its mean μ .

MA

The unconditional variance is defined as

$$\gamma_0 = E((y_t - \mu)^2)$$

Substituting the expression $y_t - \mu = u_t + \theta_1 u_{t-1}$ into the variance gives

$$\begin{aligned}\gamma_0 &= E((u_t + \theta_1 u_{t-1})^2) \\ &= E(u_t^2 + \theta_1^2 u_{t-1}^2 + 2\theta_1 u_t u_{t-1}) \\ &= E(u_t^2) + \theta_1^2 E(u_{t-1}^2) + 2\theta_1 E(u_t u_{t-1})\end{aligned}$$

MA

As u_t is independent $E(u_t u_{t-1}) = 0$, so

$$\gamma_0 = E(u_t^2) + \theta_1^2 E(u_{t-1}^2)$$

Because u_t is stationary $E(u_t^2) = E(u_{t-1}^2) = \sigma_u^2$, resulting in

$$v(y_t) \Rightarrow \gamma_0 = \sigma_u^2 + \theta_1^2 \sigma_u^2 = (1 + \theta_1^2) \sigma_u^2$$

First Order "Autocovariance"

Defined as


$$\gamma_1 = E((y_t - \mu)(y_{t-1} - \mu))$$

Substituting the expressions $y_t - \mu = u_t + \theta_1 u_{t-1}$ and $y_{t-1} - \mu = u_{t-1} + \theta_1 u_{t-2}$ into γ_1 gives

$$\gamma_1 = E((u_t + \theta_1 u_{t-1})(u_{t-1} + \theta_1 u_{t-2}))$$

$$= E(u_t u_{t-1} + \theta_1 u_t u_{t-2} + \theta_1 u_{t-1}^2 + \theta_1^2 u_{t-1} u_{t-2})$$

$$= E(u_t u_{t-1}) + \theta_1 E(u_t u_{t-2}) + \theta_1 E(u_{t-1}^2) + \theta_1^2 E(u_{t-1} u_{t-2})$$

First Order Autocovariance

As u_t is white noise $E(u_t u_{t-1}) = E(u_t u_{t-2}) = E(u_{t-1} u_{t-2}) = 0$

$$\gamma_1 = \theta_1 E(u_{t-1}^2)$$

As u_t is stationary $E(u_{t-1}^2) = \sigma_u^2$, then

$$\gamma_1 = \theta_1 \sigma_u^2 \quad = \quad E((Y_t - \mu)(Y_{t-1} - \mu))$$

$\neq 0$

Second and Higher Order Autocovariance

Defined as

$$\gamma_2 = E((y_t - \mu)(y_{t-2} - \mu))$$

Substituting the expressions $y_t - \mu = u_t + \theta_1 u_{t-1}$ and $y_{t-2} - \mu = u_{t-2} + \theta_1 u_{t-3}$ into γ_2 gives

$$\begin{aligned}\gamma_2 &= E((u_t + \theta_1 u_{t-1})(u_{t-2} + \theta_1 u_{t-3})) \\ &= E(u_t u_{t-2} + \theta_1 u_t u_{t-3} + \theta_1 u_{t-1} u_{t-2} + \theta_1^2 u_{t-1} u_{t-3}) \\ &= E(u_t u_{t-2}) + \theta_1 E(u_t u_{t-3}) + \theta_1 E(u_{t-1} u_{t-2}) + \theta_1^2 E(u_{t-1} u_{t-3}) \\ &= 0\end{aligned}$$

as u_t is independent

$$E(u_t u_{t-2}) = E(u_t u_{t-3}) = E(u_{t-1} u_{t-2}) = E(u_{t-1} u_{t-3}) = 0$$

Hence $\gamma_p = 0, p > 1$.

First Order Autocorrelation

Defined as the ratio of the first order autocovariance of y_t and the variance of y_t .

$$\rho_1 = \frac{\gamma_1}{\gamma_0}$$

For the MA(1) model this is

$$\rho_1 = \frac{\theta_1 \sigma_u^2}{(1 + \theta_1^2) \sigma_u^2} = \frac{\theta_1}{1 + \theta_1^2} \neq 0$$

Second and Higher Order Autocorrelation

Defined as

$$\rho_2 = \frac{\gamma_2}{\gamma_0} = 0$$

Hence $\rho_p = 0, p > 1$.

Estimation

Estimation of MA models requires a nonlinear algorithm as a result of the moving average component. Assuming that the disturbance term u_t is normally distributed as

$$u_t \sim N(0, \sigma_u^2)$$

the parameters

$$\Theta = \{\phi_0, \theta_1, \theta_2, \dots, \theta_q, \sigma_u^2\}$$

are estimated by maximum likelihood methods. The log-likelihood is

$$\log L = -\frac{1}{2} \log 2\pi - \frac{1}{2} \log \sigma_u^2 - \frac{1}{2\sigma_u^2 T} \sum_{t=1}^T u_t^2$$

where

$$u_t = y_t - \phi_0 - \theta_1 u_{t-1} - \theta_2 u_{t-2} - \dots - \theta_q u_{t-q}$$

Maximize $\log L$ with respect to Θ .

Forecasting

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

To forecast an MA(1) model at time $T + 1$ write the model as

$$y_{T+1} = \phi_0 + u_{T+1} + \theta_1 u_T$$

$= 0$

" t " with
" $T+1$ "

Replacing the **unknown right**-hand side terms by known values gives

$$\hat{y}_{T+1} = \hat{\phi}_0 + \hat{\theta}_1 \hat{u}_T$$

$\hat{u}_T \Rightarrow$ residual's
today.

Repeating the steps for $T + 2$ gives the forecast

$$\hat{y}_{T+2} = \hat{\phi}_0 + \hat{\theta}_1 \hat{u}_{T+1} = \hat{\phi}_0$$

$= 0$

" t " with
" $T+2$ "

as $\hat{u}_{T+1} = 0$.

Forecasting

$$\hat{\phi}_0 = \text{intercept} = 0$$
$$\hat{y}_{T+H} = 0$$

For $T + H$ the forecast is

$$\hat{y}_{T+H} = \hat{\phi}_0$$

$$\text{cos } \hat{u}_{T+H} = 0$$

with $H \geq 2$.

Notice that the forecasts from the second period onwards are all the same.

This is a property of the MA(1) model.

ARMA

The AR(p) model can be extended to allow for moving average dynamics. The ARMA(p,q) model is

$$y_t = \phi_0 + \underbrace{\phi_1 y_{t-1} + \cdots + \phi_p y_{t-p}}_{AR(p)} + u_t + \underbrace{\theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}}_{MA(q)}$$

Consider the ARMA(1,1) model

$$y_t = \phi_0 + \phi_1 y_{t-1} + u_t + \theta_1 u_{t-1}$$

The moments of the ARMA(1,1) model are summarized below

$$\text{Unconditional Mean: } \mu = \frac{\phi_0}{1 - \phi_1}$$

$$\text{Unconditional Variance: } \gamma_0 = (1 + \theta_1^2 + 2\phi_1\theta_1) \frac{\sigma_u^2}{1 - \phi_1^2}$$

$$\text{First order autocovariance: } \gamma_1 = \frac{(1 + \phi_1\theta_1)(\phi_1 + \theta_1)\sigma_u^2}{1 - \phi_1^2}$$

$$\text{First order autocorrelation: } \rho_1 = \frac{(1 + \phi_1\theta_1)(\phi_1 + \theta_1)}{1 + \theta_1^2 + 2\phi_1\theta_1}$$

$$\text{Higher order autocorrelation: } \rho_p = \phi_1 \rho_{p-1}, \quad p \geq 2$$

Application

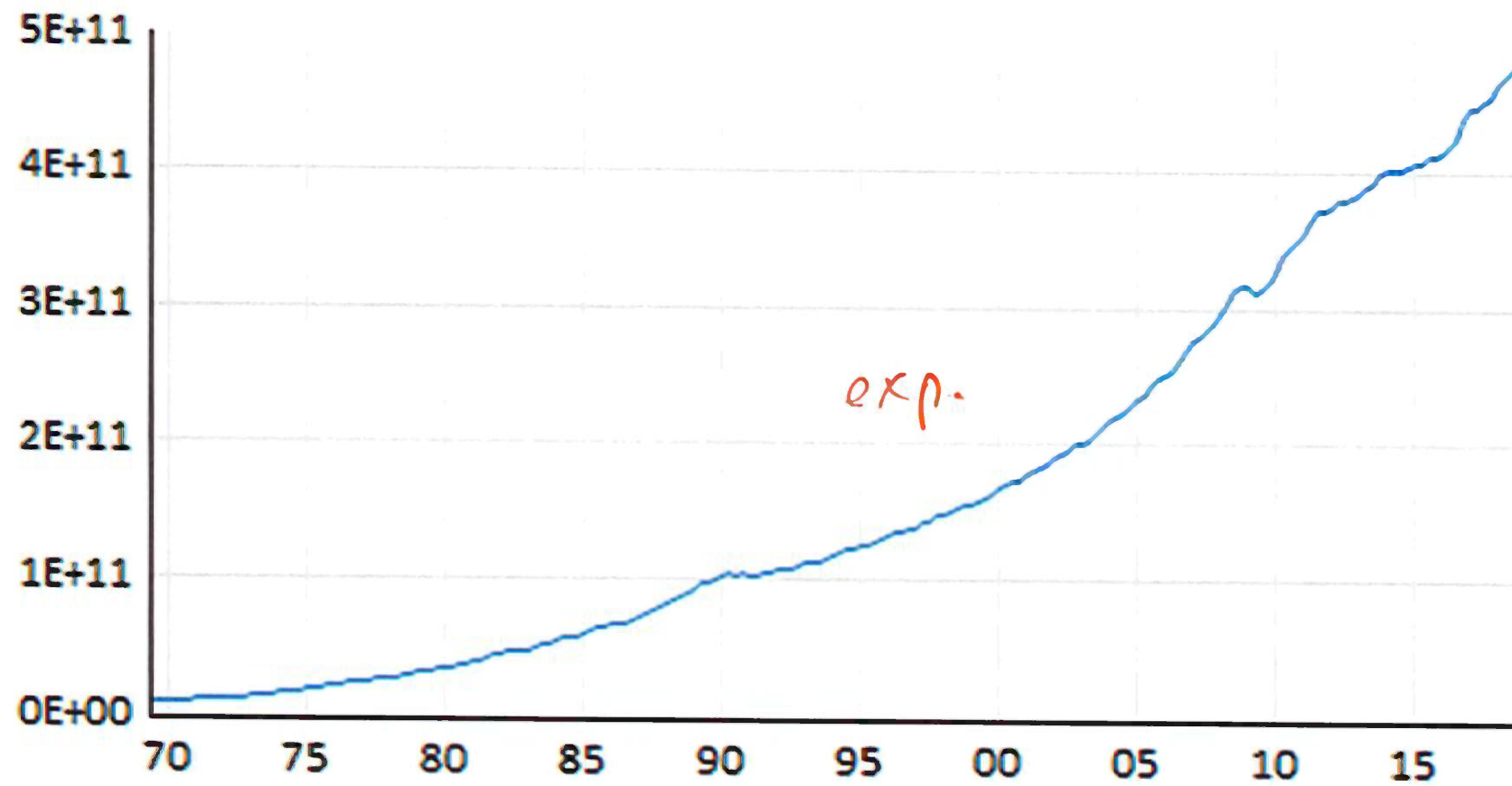
You worked at a Reserve Bank and were tasked with forecasting the GDP growth rate.

The following figure gives time series plots of 2 time series for Australia from 1969Q3 to 2019Q1.

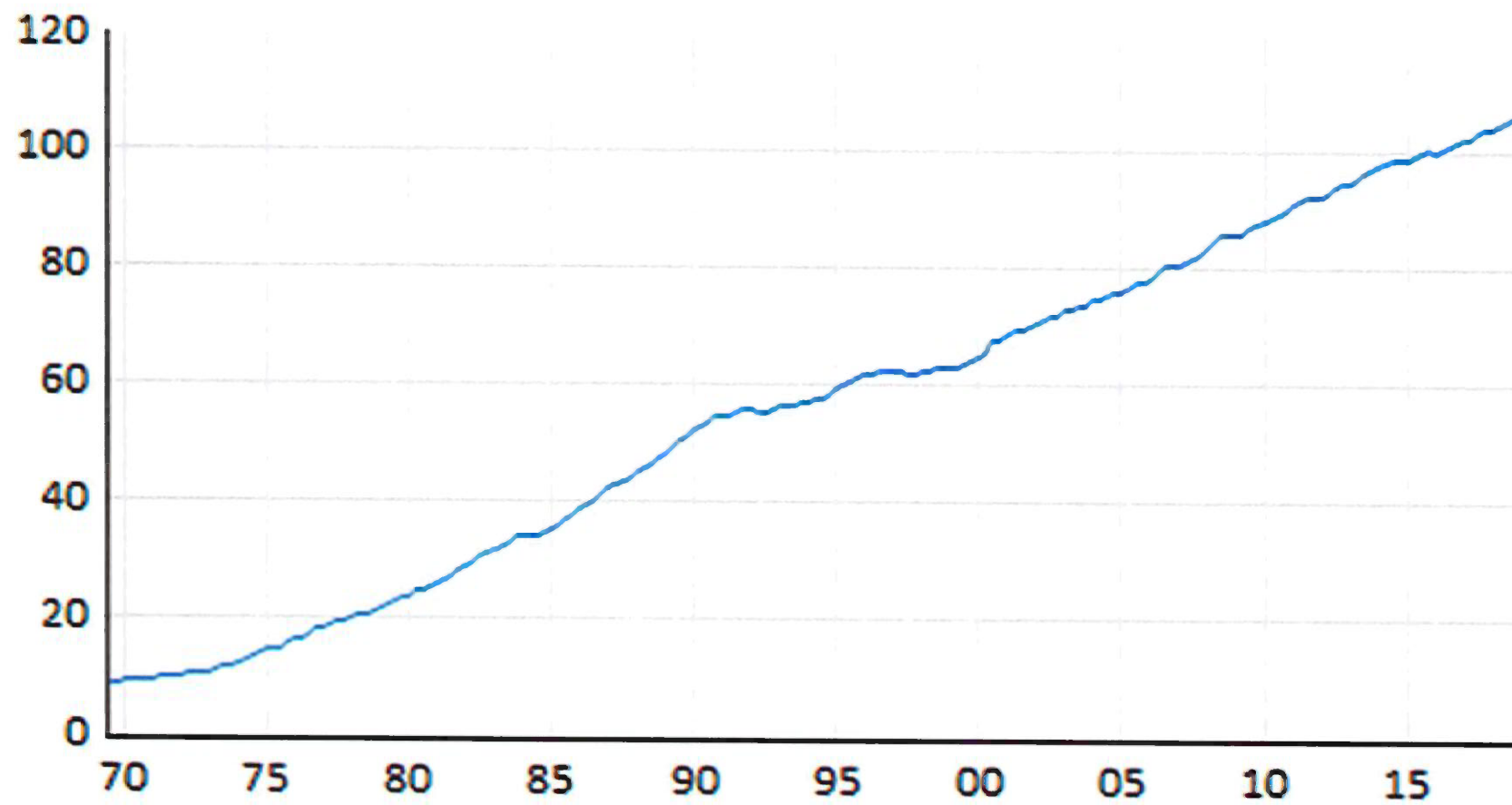
NGDP - Nominal GDP, seasonally adjusted 💵

PRICE - Nominal price (CPI), seasonally adjusted

Nominal GDP



PRICE (CPI)



Real GDP

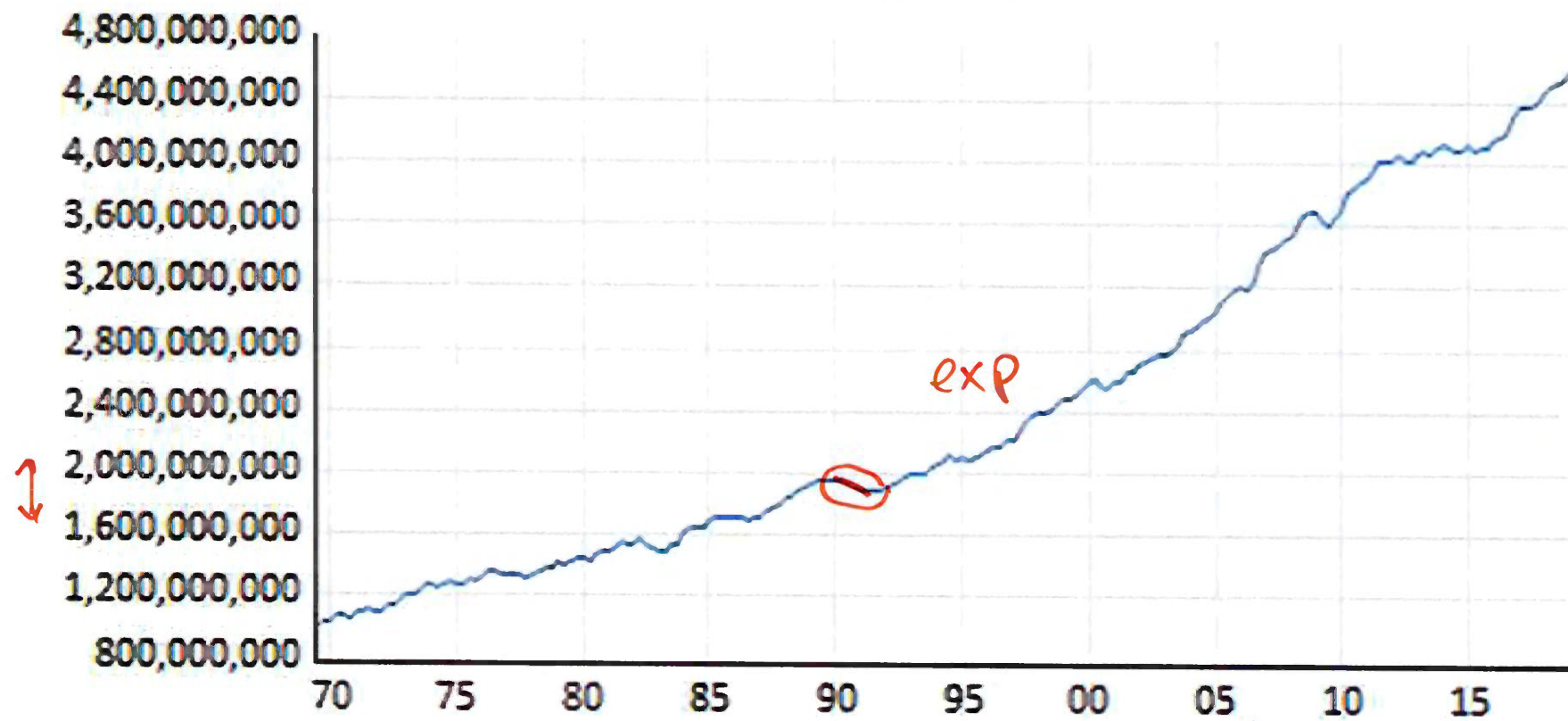
not fake

Real GDP is defined as the ratio of nominal GDP (NGDP) and the price level (PRICE)

$$RGDP = \frac{NGDP}{PRICE}$$

Graph shows that RGDP and NGDP both trend upwards with NGDP exhibiting a relatively smoother trend.

Real GDP



Transformation

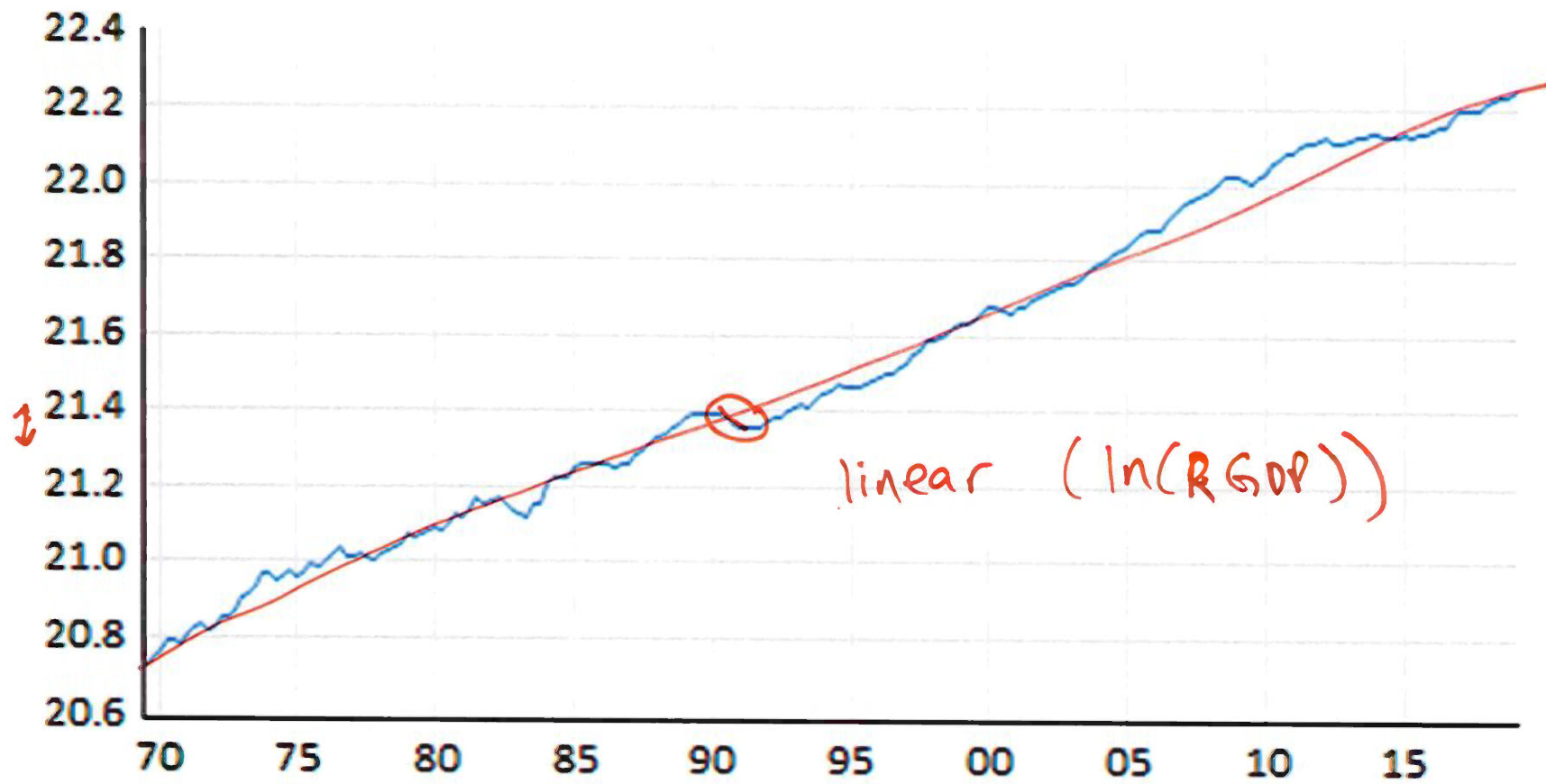
The transformation is $\log(y_t)$.

The graph shows that RGDP has an increasing trend whereas LRGDP has a linear trend.

Log filter has two important characteristics:

- ▶ Capture nonlinearities
- ▶ Can correct for changing volatility (big movements are dampened more than smaller movements)

LN_RGDP



Growth Rate

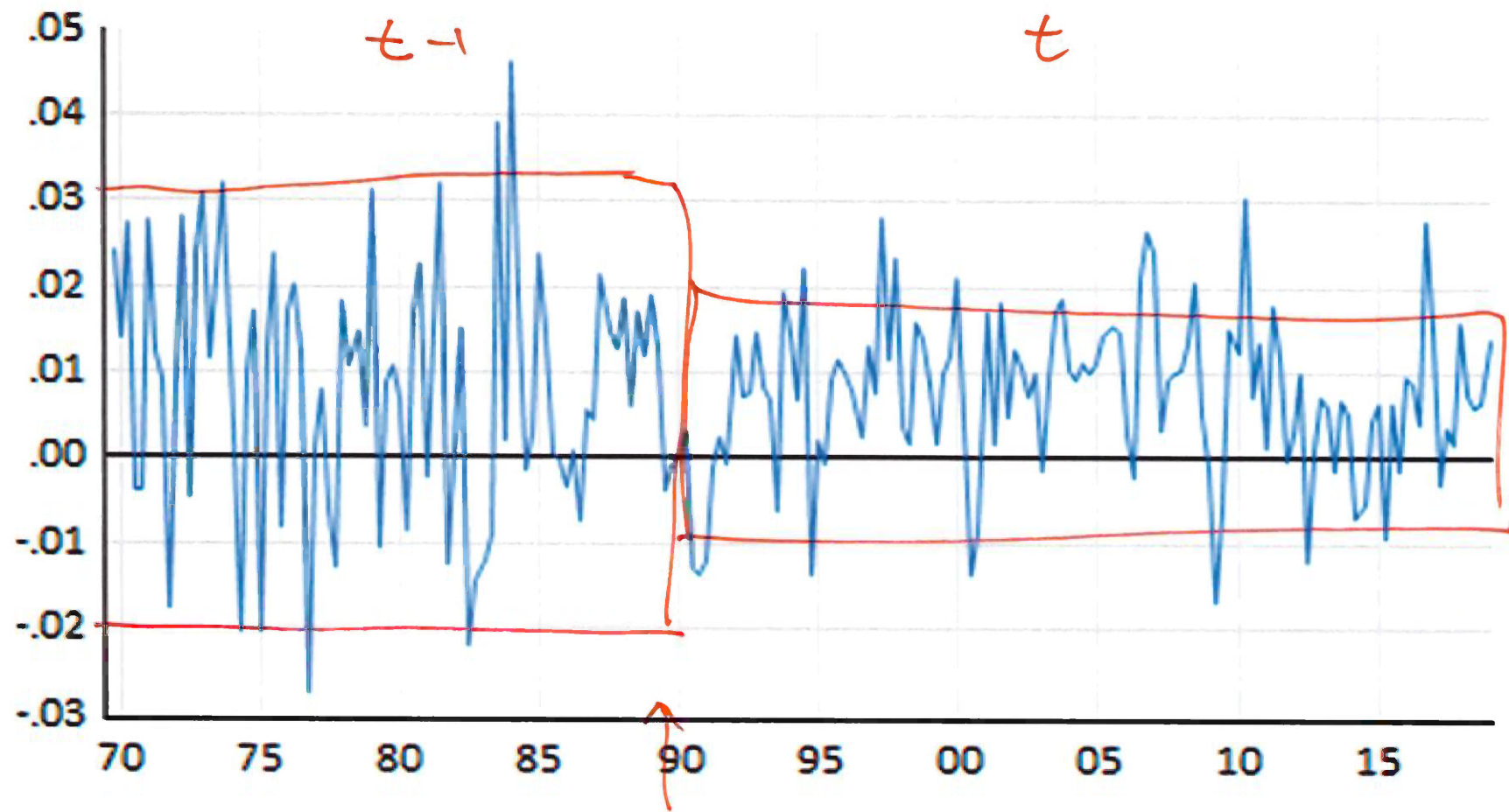
The aim of this application is to specify an ARMA model to forecast the growth rate of real GDP in Australia from 2019Q2 to 2020Q4.

The growth rate is defined as

$$DLRGDP_t = \log(RGDP_t) - \log(RGDP_{t-1})$$

This series is plotted in the next slide.

GROWTH_RATE



$v(y_t) = v(y_{t-1}) \Rightarrow$ violate stationary condition.

$\begin{matrix} t-1 & | & t \\ \hline \text{[wavy line]} & & \text{[wavy line]} \end{matrix}$

```
df <- readxl::read_excel("week 11_rba.xlsx")
```

```
library(xts)
```

```
n <- nrow(df)
```

```
idx_q <- as.Date(as.yearqtr("1969 Q3") + (0:(n - 1)) / 4, frac = 1)
```

```
data.xts <- xts(df, order.by = idx_q)
```

```
index(data.xts) <- as.yearqtr(index(data.xts))
```

```
plot(data.xts$NGDP) slide 53
```

```
plot(data.xts$PRICE) " 54
```

real $\text{rgdp} = \text{data.xts\$NGDP} / \text{data.xts\$PRICE} \Rightarrow \text{slide 55}$

↓
nominal

```
# growth rate
```

```
growth_rate <- na.omit(diff(log(rgdp))) ⇒ slide 60
```

ARMA

The results from estimating the ARMA(p,q) for alternative combinations of p and q are given on the next few slides.

base R x-ts AR(1)

↑ ↓ ↑

```
> # AR(1)
> fit_ar1 <- arima(as.numeric(growth_rate), order = c(1,0,0), include.mean = TRUE)
> fit_ar1
```

call:
 arima(x = as.numeric(growth_rate), order = c(1, 0, 0), include.mean = TRUE)

Coefficients:

	ar1	intercept
	0.1107	0.0077
s.e.	0.0708	0.0010

sigma^2 estimated as 0.000141: log likelihood = 596.82, aic = -1187.64

$$y_t - \mu = \phi_1 (y_{t-1} - \mu) + \epsilon_t$$

↓

$\hat{\mu}$

AR(2)

↓

```
> fit_ar2 <- arima(as.numeric(growth_rate), order = c(2,0,0), include.mean = TRUE)
```

```
> fit_ar2
```

call:

```
arima(x = as.numeric(growth_rate), order = c(2, 0, 0), include.mean = TRUE)
```

Coefficients:

	ar1	ar2	intercept
	0.1141	-0.0319	0.0077
s.e.	0.0712	0.0711	0.0009

sigma^2 estimated as 0.0001409: log likelihood = 596.92, aic = -1185.84

AR(1) MA(1)
↓ /

```
> fit_arma11 <- arima(as.numeric(growth_rate), order = c(1,0,1), include.mean = TRUE)
> fit_arma11

call:
arima(x = as.numeric(growth_rate), order = c(1, 0, 1), include.mean = TRUE)

Coefficients:
          ar1          ma1      intercept
      -0.4381    0.5570         0.0077
s.e.    0.3652    0.3364         0.0009
```

ARMA(1,1)

sigma^2 estimated as 0.0001403: log likelihood = 597.31, aic = -1186.63

```
> fit_ma1 <- arima(as.numeric(growth_rate), order = c(0,0,1), include.mean = TRUE)
```

```
> fit_ma1
```

AR(1)

MA(1)

call:

```
arima(x = as.numeric(growth_rate), order = c(0, 0, 1), include.mean = TRUE)
```

Coefficients:

	ma1	intercept
	0.1213	0.0077
s.e.	0.0750	0.0009

```
sigma^2 estimated as 0.0001409: log likelihood = 596.92, aic = -1187.84
```

```
> res.ma1 <- residuals(fit_ma1) ⇒ numeric (base R)  
> res.ma1.xts <- xts(res.ma1, order.by = index(growth_rate))  
> tail(res.ma1.xts)
```

```
      [,1]  
2017 Q4 -0.005610346  
2018 Q1  0.008858538  
2018 Q2 -0.001146712  
2018 Q3 -0.001418741  
2018 Q4 -0.001092497  
2019 Q1  0.006526154 =  $\hat{u}_T$ 
```

↓
print data

MA(2)

```
> fit_ma2 <- arima(as.numeric(growth_rate), order = c(0,0,2), include.mean = TRUE)
> fit_ma2
```

Call:
arima(x = as.numeric(growth_rate), order = c(0, 0, 2), include.mean = TRUE)

Coefficients:

	ma1	ma2	intercept
	0.1235	-0.0405	0.0077
s.e.	0.0714	0.0688	0.0009

sigma^2 estimated as 0.0001406: log likelihood = 597.1, aic = -1186.19

MA(1)

Let y_t denote the growth rate of real GDP (DLRGDP).

→ slide 66

Since lower AIC is better, and -1187.84 is the smallest value, the MA(1) model is preferred by AIC among the models considered.

We consider an MA(1) model to generate forecasts.

$$y_t = \phi_0 + u_t + \theta_1 u_{t-1}$$

$\boxed{\text{MA}(1)}$
 $E(y_t) = \phi_0$
= intercept.

The fitted MA(1) model

↑ intercept ma1

$$y_t = 0.007726 + \hat{u}_t + 0.121308\hat{u}_{t-1}$$

where $\hat{u}_t$ is the residual with variance $\hat{\sigma}_u^2 = 0.000141$.

↑

$\boxed{\text{AR}(1)}$
 $E(y_t) = \frac{c}{1 - \phi_1}$

Forecast

To forecast an MA(1) model at time $T + 1$, we have $\hat{u}_T = 0.0065$ and so

$$\begin{aligned}\hat{y}_{T+1} &= \hat{\phi}_0 + \hat{\theta}_1 \hat{u}_T \\ &= 0.007726 + 0.121308 (0.006526) \\ &= 0.0085177\end{aligned}$$

↓
slide 66

At time $T + 2$,

$$\begin{aligned}\hat{y}_{T+2} &= \hat{\phi}_0 \\ &= 0.007726\end{aligned}$$

MA(1)

The results of forecasting the MA(1) model from 2019Q2 to 2020Q4 are given below.

growth rate
real

	Quarter	DLRGDP	Forecast
last obs $\Rightarrow$	2019Q1	0.014120	0.014120
forecast $\Rightarrow$	2019Q2	NA	0.008518
	2019Q3	NA	0.007726
	2019Q4	NA	0.007726
	2020Q1	NA	0.007726
	2020Q2	NA	0.007726
	2020Q3	NA	0.007726
	2020Q4	NA	0.007726

$\hat{Y}_{T+1}$
 $\hat{Y}_{T+2}$
 $\hat{\phi}_0$

Forecasts

The forecasts converge to a quarterly growth rate of 0.007726. This represents an annualized growth rate of

$$4 \times 0.007726 = 0.030904$$

or 3.09% per annum.

5% p.a.

The following table contains the IMF projections of real GDP (annual percentage growth rates) for 2021 and 2022.

Latest World Economic Outlook Growth Projections

(real GDP, annual percent change)	2020	PROJECTIONS	
		2021	2022
World Output	-3.1	5.9	4.9
Advanced Economies	-4.5	5.2	4.5
United States	-3.4	6.0	5.2
Euro Area	-6.3	5.0	4.3
Germany	-4.6	3.1	4.6
France	-8.0	6.3	3.9
Italy	-8.9	5.8	4.2
Spain	-10.8	5.7	6.4
Japan	-4.6	2.4	3.2
United Kingdom	-9.8	6.8	5.0
Canada	-5.3	5.7	4.9
Other Advanced Economies	-1.9	4.6	3.7

Source: World Economic Outlook, October 2021